Language Models Learn from Medical Textbooks

AI Research Paper Explainer

Introduction

In the research paper *Small language models learn enhanced reasoning skills from medical textbooks*, Kim et al. (2025) introduce Meerkat, a new family of medical small language models (SLMs) designed to be lightweight while enhancing reasoning capabilities, along with a training method that can be used to enable reasoning in SLMs.This work addresses a fundamental challenge: while Chain of Thought (CoT) prompting has been one of the largest advancements for large language models (LLMs) in recent memory, initial findings showed that these improvements do not manifest in SLMs as they do with LLMs, with results showing that CoT prompting even reduces the accuracy of models with fewer than 10 billion parameters.

CoT prompting demonstrated that LLMs can be taught not only to chase the most probable next token, but to have in mind a reasoning task that it breaks down into a series of intermediate steps and tries to solve. This led to a significant increase in accuracy across commonsense, symbolic, and mathematical reasoning tasks. Initially, this breakthrough was credited to abilities such as semantic understanding and symbolic mapping, which were thought to emerge only at larger scales, leaving SLMs overlooked.

Over the past two years, the glass ceiling for SLM reasoning has begun to show cracks, with newer models trained through distillation, which demonstrated gains in accuracy compared to their instruction model counterparts. However, these improvements remain insufficient for domains like healthcare, which would greatly benefit from efficient and rapid decision-making AI assistants. Although LLMs could meet this demand, adoption has been limited by patient privacy concerns. In contrast, SLMs can be easily hosted on commercially available hardware that healthcare institutions can readily acquire and maintain.

This need in the healthcare industry drove the authors to propose a practical middle path: improve reasoning in SLMs by fine-tuning them on high-quality chain-of-thought (CoT) reasoning examples derived from medical textbooks and questions from the United States Medical Licensing Examination (USMLE). Their approach consists of three parts: (1) extracting detailed stepwise rationales from a large-capacity LLM (GPT-4) for curated questions, (2) synthesizing additional CoT examples from textbook content, and (3) instruction-tuning SLM backbones (Mistral-7B and Llama-3-8B) on the combined dataset. The result is Meerkat-7B and Meerkat-8B, which achieve state-of-the-art performance among open-weight SLMs on multiple medical benchmarks and perform competitively with much larger commercial models on case challenges.

Methodology

Data Sources

The authors used multiple-choice USMLE-style questions from the MedQA dataset, alongside GPT-4 to generate stepwise rationales for each MedQA item. MedQA is a public medical question-answering dataset made from both real and practice medical board exam questions used to evaluate clinical reasoning in NLP models. GPT-4 was prompted to analyze the vignette and options step-by-step and to produce a formatted CoT explanation and final choice. Only GPT-4 outputs that followed the required output template and whose

final answer matched the gold answer were retained. After filtering, 9.3K out of 10K MedQA items remained and formed MedQA-CoT.

MedQA-CoT was further augmented with CoT examples generated using medical textbooks. The authors chunked 18 medical textbooks spanning core disciplines and prompted GPT-4 to synthesize grounded multiple-choice and free-response QA pairs with CoT rationales from those chunks, producing around 78K additional CoT examples. This portion of the data was named Textbook-CoT.

Textbook CoT and MedQA CoT were merged with multiple medical instruction datasets (MedMCQA, MedicationQA, LiveQA, ChatDoctor-cleaned, MedInstruct-52K, MedQA-dialog, etc.) to create a multi-task corpus of around 441K examples covering multiple-choice, free-form QA, multi-turn dialogue, and clinical-note generation.

Model Training

The Meerkat 7 billion parameter model (Meerkat-7B) is initialized from the Mistral-7B-0.1 foundation model, and the Meerkat 8 billion parameter model was initialized from Meta's Llama-3-8B-Instruct. Both models used a standard autoregressive next-token prediction objective, with Meerkat-7B training for one day, in which it completed three epochs with batch size 128, learning rate 2e-6, and maximum sequence length 2,048 tokens, using eight 80GB A100 GPUs. Meerkat shared most of the training setup, with the only difference being a higher learning rate of 7e-6 and a Google TPU setup instead of the Nvidia GPUs. The authors report that the CoT fine-tuning pipeline used ~28× less GPU time than comparable continuous pretraining approaches for 7B models.

Evaluation

Quantitative evaluation was conducted using the MedQA, USMLE sample test, Medbullets-4, Medbullets-5, MedMCQA, and MMLU-Medical benchmarks, along with the NEJM Case Challenges (38 cases). The benchmarks were used to run a controlled ablation study to better isolate the effects of training on QA only (MedQA), training on MedQA + MedQA-CoT, and training on MedQA-CoT + Textbook-CoT. The aforementioned evaluations were done for two model settings: single model response, and an ensemble response, where the model is prompted to give multiple answers, and a majority vote is taken

Moreover, Qualitative Human A/B testing was conducted on Meerkat-8B and Llama-3-8B-Instruct using 50 selected samples from the Medbullets-5 benchmark (out of 101 total samples where models answered correctly). Ten experts (4 medical students + 6 MD physicians) participated; they were split into two groups of 5 based on experience/licensure; each group evaluated the same 25 questions, and final judgment per question was by majority vote. Evaluators could choose the better model or a draw based on the following metrics: completeness, factuality, clarity, and logical consistency.

Takeaways

Quantitative Results

Meerkat-7B and Meerkat-8B achieved an average accuracy of ≈64.5% and ≈66.7% across the six medical benchmarks, which is significantly higher than the foundation models they were based on (e.g., Mistral-7B ≈41.2%, Llama-3-8B ≈56.1%). On MedQA specifically, Meerkat-7B achieved ≈71.2% (single model) and ≈77.1% (ensemble), exceeding common USMLE passing thresholds of approximately 60%. The ablation studies shed more light on the reasons behind the gains in accuracy, with CoT fine-tuning on MedQA contributing +7.5% accuracy, and the addition of textbook-derived CoT yielding an additional +5.1% improvement.

Both models improved on the NEJM case challenge scores of their foundation models, with Meerkat-7B and Meerkat-8B scoring 16 and 20, compared to 7 and 13 for Mistral-7B-0.1 and Llama-3-8B-Instruct. The score achieved by Meerkat-8B is especially significant as it surpasses the human score of 13.7 and closes in on GPT-4's score of 21.8.

Qualitative Results

Clinician A/B tests favored Meerkat-8B over baseline SLMs for completeness and logical consistency. Meerkat tends to produce longer, more thorough rationales, which cover more relevant differential diagnoses and decision-points than the Llama-3-8B baseline. Moreover, raters noted that the Textbook-derived CoT examples reduced blatant factual drift in many rationales; the model more often cited mechanisms, timelines, or guideline-style reasoning rather than speculative text.

In spite of the high praise Meerkat-8B received compared to its Llama backbone, the model still exhibited some limitations and failures. Meerkat still produced incorrect clinical facts on occasion (e.g., dosage details, absolute contraindication misstatements). The authors explicitly warn about these safety risks. Additionally, Textbook-CoT did not eliminate all hallucinations; when the training chunk did not fully cover a subtopic, the generator sometimes filled gaps with plausible-sounding but unsupported details. Lastly, human raters noted that the increased verbosity of Meerkat-8B lowered perceived clarity as it occasionally impaired quick readability for clinicians. Raters preferred concise, high-signal explanations in some clinical contexts.

Application

The Meerkat family of models showcases a modern example of AI applications in healthcare that could significantly impact public health. As demonstrated, these models outperform humans in certain case detection scenarios, such as those in the NEJM challenges (benchmark tasks introduced by the New England Journal of Medicine). Any statistically significant improvement in this area represents not just incremental progress but, more importantly, a life saved. Even so, adoption of AI-based systems in medicine remains limited. Privacy concerns have largely been addressed by systems like Meerkat, yet model hallucinations are still viewed as too great a threat to public safety, even when largely mitigated. Another concern is the risk of over-reliance on AI, which many fear could erode physicians' ability to practice their craft. Medicine requires years of studying vast amounts of literature, but it is equally grounded in hands-on training, practical experience, and the capacity to make critical decisions in real time. Many worry that these essential skills will diminish if decision-making is increasingly delegated to AI systems. Still, as AI technology matures and regulators gain a deeper understanding of its strengths and limitations, models like Meerkat may find a place in medicine; not as replacements for physicians, but as powerful tools that responsibly enhance clinical practice and improve patient outcomes.

References

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